Openhole completion integrity analysis for life of CO_2 injector well

M Rashad Amir Rashidi, Tan Chee Phuat, M Hamzi Yakup, Ismatul Hani Bt Idris, Muhamad Nasrul Amri Muhamad Zaidi, Latief Riyanto, M Nizam B M Zin, M Azuan B Abu Bakar*
PETRONAS

Copyright 2024 ARMA, American Rock Mechanics Association
This paper was prepared for presentation at the International Geomechanics Conference (IGS) on 18 -20 November 2024 in Kuala Lumpur, Malaysia. This paper was selected for presentation at the conference by an IGS Technical Program Committee based on a technical and critical review of the paper by a minimum of two technical reviewers. The material, as presented, does not necessarily reflect any position of the partner societies ARMA/DGS/SEG/SPE/AAPG/SPWLA/CSRME/EAGE/SRMEG/MOGSC, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper for commercial purposes without the written consent of the partner societies is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 200 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgement of where and by whom the paper was presented.

ABSTRACT: A study was conducted to investigate the openhole completion integrity for a CO_2 injector well in a Carbon Capture and Storage (CCS) project in offshore Sarawak, Malaysia. A number of completion designs were considered, including openhole completions. Hence, the main objective of this study is to analyse the stability of the openhole section from pre-CO_2 injection until end of injection. As part of the project, one-way coupled thermal geomechanics-dynamic modelling was conducted to assess the geomechanical containment risks. Based on the coupled modelling results, significant stress changes are observed at the injector during the injection period due to the effects of injection and CO_2 cooling which increased and decreased the stresses, respectively. The stresses, temperatures and pore pressures were then extracted from the coupled model at several key time-steps throughout the injection period and used for openhole stability analysis. Several scenarios were analysed which correspond to the end of drilling, after drilling fluid is displaced with brine prior to injection, and start, intermediate and end of the CO_2 injection. Additionally, the stability analysis was also conducted for scenario whereby injection is paused, a scenario which may potentially occur due to operational-related issues. The potential integrity risk of the openhole is determined based on the magnitude of wellbore failures predicted in the analysis. The results of the openhole stability analysis of the injector well throughout CO_2 injection showed that minor to considerable shear failures and hydraulic fractures are predicted, in particular during injection stoppages. During the stoppages, the predicted breakouts become more severe at specific sections within the reservoir formation. As there are potential risks of openhole instability during the life of CO_2 injection, perforated liner is adopted as the completions for the reservoir interval. Similar analysis is recommended for future CCS projects to ensure the completions method is aligned with the overarching goal of ensuring safe, long-term CO_2 storage in the reservoir, protecting both the environment and the integrity of the overall storage project.

1. INTRODUCTION

Carbon Capture and Storage (CCS) has emerged as one of the key methods in combating climate change. The International Energy Agency (IEA) estimates that global carbon dioxide emissions from the energy sector have reached a new record high of 37 billion tonnes in 2022 and is set to peak within the decade (International Energy Agency, 2023). This has prompted a global effort towards reducing the global carbon footprint which have been led by various countries and institutions around the globe. Many organizations have committed to achieving net zero carbon emissions by 2050, recognizing CCS as a critical component of this effort (Global CCS Institute, 2022). CCS, which is the process of capturing large amounts of CO_2 from CO_2-contaminated hydrocarbon and industrial sources and injecting them into subsurface reservoirs for permanent storage (Metz et al., 2005), has been identified as one of the key strategies in minimizing the organization and country carbon footprint. However, there are various risks involved in CCS operations including CO_2 leakage, fault reactivation and induced seismicity, injectivity issues, and wellbore integrity breach, which underscores the complexity and importance of thorough risk assessment throughout the project lifecycle (Rutqvist and Tsang, 2002 and Rutqvist et al., 2009).

This study focuses on the investigation of openhole completion integrity for a CO_2 injector well in a CCS project in offshore Sarawak, Malaysia. An important aspect of designing a well is the selection of the completions strategy. Various options exist, including cased-hole completion, openhole completion and open hole with perforated liners, with each options offering unique advantages and challenges, especially in the context of CO_2 storage. Openhole completions in particular, offer possible cost savings and potentially better injectivity, but they also present risks to wellbore stability and integrity, especially in challenging offshore environments where stress-induced failures, temperature changes, and chemical interactions with CO_2 can exacerbate wellbore instability (Fjaer et al., 2008 and Cappa and Rutqvist, 2011). Hence, detailed analysis is required to analyse the stability of the openhole section from pre-CO_2 injection until end of injection. As part of the overall project, one-way coupled thermal geomechanics-dynamic modelling was conducted to assess the geomechanical containment risks. The simulation conducted uses a thermo-hydro-mechanical

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026

(THM) framework, integrating the interactions between temperature, fluid flow, and mechanical stresses. The thermal model incorporated the relevant heat transfer mechanisms such as advective and diffusive heat transport and where the thermoelastic effects on stresses and pore pressures were modeled to understand how temperature changes impact the geomechanical behaviour of the reservoir and caprock. This is particularly relevant in CO_2 injection scenarios, where dynamic changes in pressure and temperature can leave the wellbore exposed to stress-induced failures (Rutqvist and Tsang, 2002). During CO_2 injection, the injected CO_2 is typically cooler than the reservoir temperature (Pruess, K., 2008). Therefore, the CO_2 will be reaching the target formation at a lower temperature (Paterson et al., 2008) as it does not have sufficient time to thermally equilibrate when it flows through the injector well. The large temperature difference can lead to thermal stresses which may induce changes in pore pressures, effective stresses (Vilarrasa et al., 2015) and potentially to the mechanical properties of the rocks.

Based on the coupled modelling results, significant stress changes are observed at the injector during the injection period. The effects of CO_2 injection contributed to the increase in reservoir pressure, thereby increasing the horizontal stresses, while the cooling effects due to the temperature difference between the injected CO_2 and reservoir temperature caused decrease in the horizontal stresses (Cappa and Rutqvist, 2011 and Vilarrasa et al., 2014). These two opposing effects observed in the coupled modelling results were then considered in the wellbore stability analysis to understand the potential impact on the stability of openhole completions in the injector well.

In addition to the continuous injection scenario, this study also considers the impact of wellbore stability during injection stoppages, which may occur due to operational issues or interruptions. During these stoppages, stress redistribution may occur, leading to shear failures or breakouts. Such failures could compromise the wellbore ability to contain the injected CO_2, posing significant risks to the long-term storage objectives of the project. The evaluation of these scenarios are conducted to assess the potential impacts on the injector well integrity (Paterson et al., 2008).

2. METHODOLOGY

The openhole stability analysis is conducted by extracting the stresses and pore pressures from the one-way coupled thermal geomechanics-dynamic model at key time-steps throughout the injection period. The four key time-steps identified were:

- Year 2025: Post drilling and prior to injection
- Year 2030: 5 years after injection
- Year 2040: 15 years after injection
- Year 2048: 23 years after injection

Large horizontal stress reduction due to CO_2 cooling (stress decrease) and pressure increase (stress increase) were observed during injection period at all injector wells. However, injector well I-3 (Fig 1) is selected for the openhole stability analysis as the well is identified to be least stable (marginally) during drilling scenario.

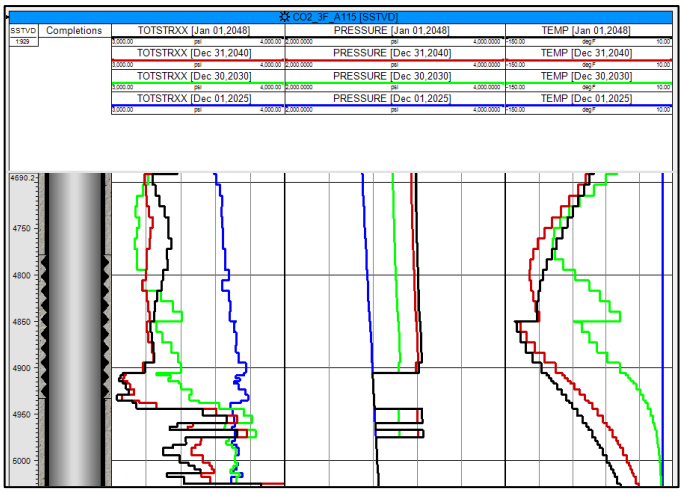


Fig 1: Injector I-3 showing the temperature (right), pressure (middle) and total minimum horizontal stress variation (left) throughout the key time-steps (blue: Y2025, green: Y2030, maroon: Y2040, black: Y2048).

The maximum temperature and minimum horizontal stress reductions are observed to occur in the final timestep (Y2048) as observed in Fig 1.

For the openhole stability analysis, several scenarios were considered for evaluation which included:

- End of drilling: Assessing wellbore stability immediately following the end of drilling process.
- After drilling mud is displaced with brine: Evaluation after the drilling mud is displaced with brine, prior to CO_2 injection.
- During CO_2 injection: Assessing wellbore stability during CO_2 injection stage.
- Upon injection stoppage: Assessing wellbore stability when the injection process is paused due to operational issues and interruptions.

These different scenarios represent the possible circumstances which the wellbore may experience throughout the CO_2 storage lifecycle. As observed in Fig 1, one of the main effects of CO_2 injection is the thermal cooling experienced at the injector due to the lower temperature CO_2 being injected into the higher temperature reservoir. The cooling, which can reach as much as 139°F, greatly reduces the stresses, thereby

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026

reducing the fracture gradient of the reservoir. This can contribute to hydraulic fracturing during the injection process. Meanwhile, during injection stoppages, other issues may arise such as the horizontal stresses exceeding the wellbore pressures, causing shear failures or breakouts. Hence, the potential integrity risk of the openhole is determined based on the magnitude and severity of breakouts and other shear failure modes, as well as hydraulic fractures predicted in the analysis during the different scenarios. Typical wellbore failures which include breakouts and fractures can be observed in Fig 2.

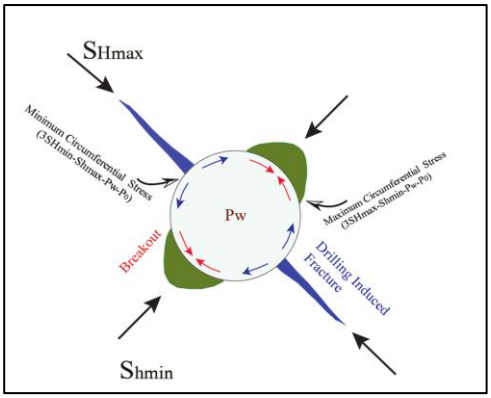


Fig 2: Typical borehole breakout and drilling-induced fractures which may cause wellbore stability issues (Fjaer et al., 1992).

3. OBSERVATION AND RESULTS

The first scenario evaluated is as shown in Fig 3, where the reservoir section is shown to be drilled using 11.3 ppg mudweight. Based on the wellbore stability analysis, there are minor breakouts predicted within the reservoir interval.

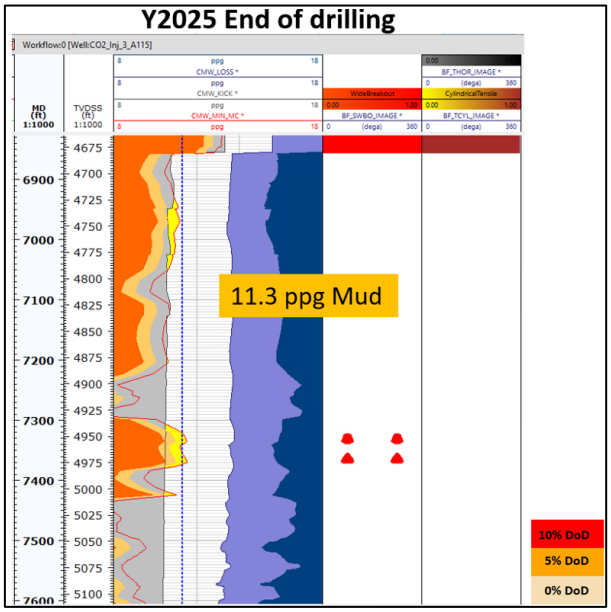


Fig 3: Injector I-3 in Y2025 at the end of drilling, where the well is drilled with 11.3 ppg mud weight. Minor breakouts predicted at ~7330 ftMD and 7360 ftMD. DoD defined as the "depth of damage".

The evaluation is then followed by the second scenario where the drilling mud is displaced with 12.3 ppg brine, as shown in Fig 4. During this stage, the wellbore stability analysis showed a stable wellbore with no occurrence of breakouts or hydraulic fractures.

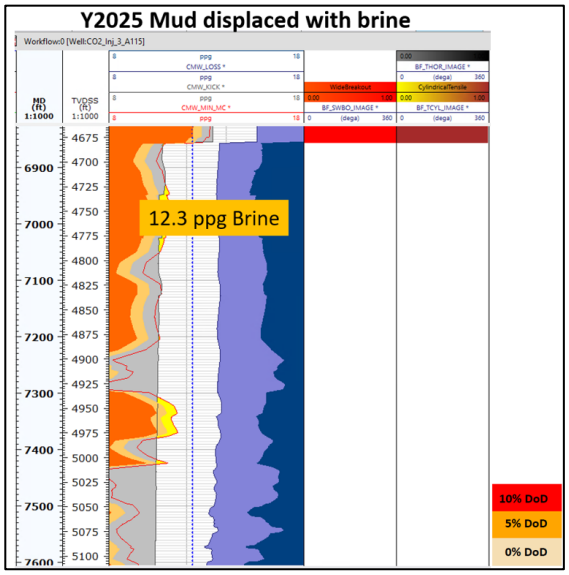


Fig 4: Injector I-3 in Y2025 where drilling mud is displaced with 12.3 ppg brine. No wellbore failure is predicted. DoD defined as the "depth of damage".

Meanwhile, Fig 5 and Fig 6 present the results of the wellbore stability analysis, showing scenarios during CO_2 injection and after injection stoppage, respectively. During the CO_2 injection at 13.8 ppg, the wellbore did not exhibit any failures, whereas during the injection stoppage with 10.8 ppg reservoir pore pressure, considerable breakouts were predicted at certain reservoir sections.

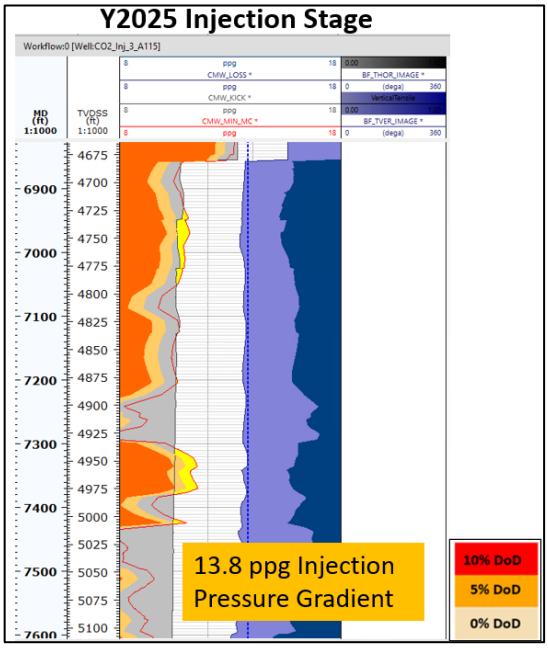


Fig 5: Injector I-3 in Y2025 with CO_2 injection at 13.8 ppg pressure gradient (left). No breakout predicted during continuous injection. DoD defined as the "depth of damage".

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026

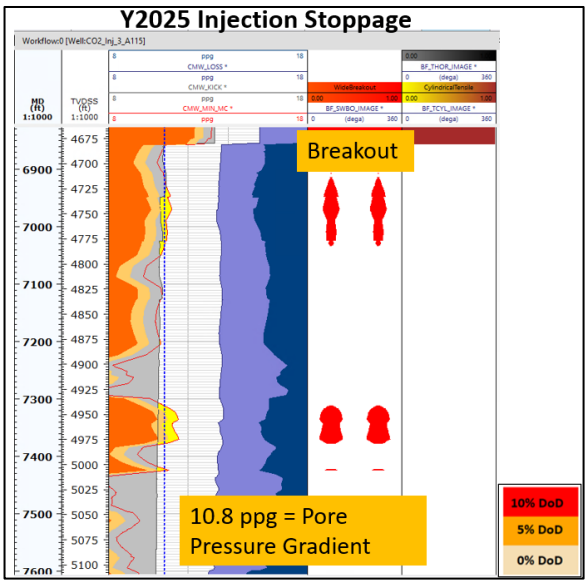


Fig 6: Injector I-3 in Y2025 during injection stoppage. Wide breakouts predicted at ~6920-7030 ft MD and ~7315-7375 ft MD during injection stoppage. DoD defined as the "depth of damage".

After 5 years of injection, the prediction of the wellbore stability is as shown in Fig 7. During Y2030 with 12.3 ppg injection pressure, considerable hydraulic fractures are predicted in the shallower section of the reservoir at ~7000-7220 ft MD. In the event of injection stoppage, with the reservoir pore pressure at 11.75 ppg, minor breakout and hydraulic fractures are predicted as observed in Fig 8.

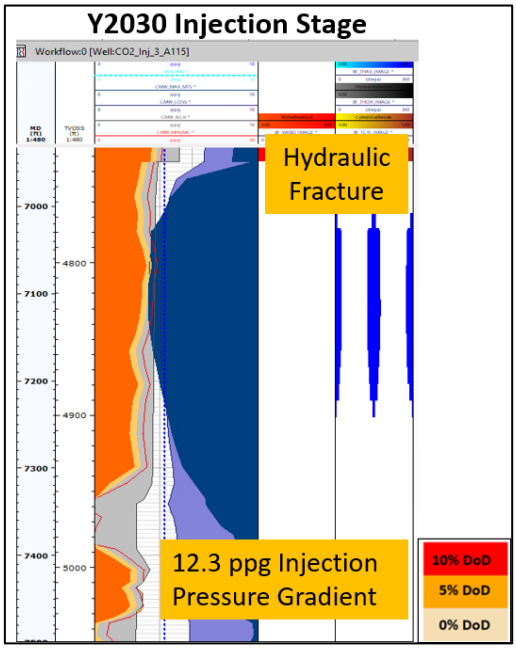


Fig 7: Injector I-3 in Y2030 during continuous CO_2 injection at 12.3 ppg pressure gradient. Hydraulic fracture predicted at ~7000-7220 ft MD during the injection. DoD defined as the "depth of damage".

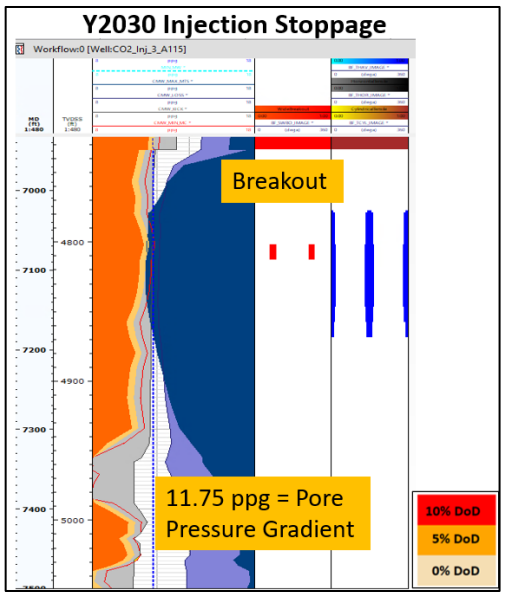


Fig 8: Injector I-3 in Y2030 during injection stoppage with minor breakout at ~7070-7085 ft MD and hydraulic fractures at ~7020-7170 ft MD were predicted. DoD defined as the "depth of damage".

In Y2048, the predicted wellbore stability analysis is as shown in Fig 9. During CO_2 injection with 13.0 ppg, considerable shear failures (narrow breakout) and hydraulic fractures are predicted along the reservoir interval. Upon injection stoppage with 12.45 ppg reservoir pore pressure, considerable wide breakouts are predicted in the shallower section, whereas hydraulic fractures are predicted in the middle section of the reservoir, as observed in Fig 10.

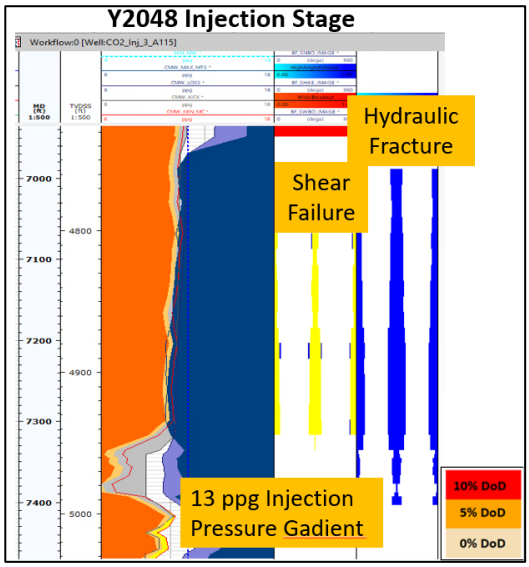


Fig 9: Injector I-3 in Y2048 during continuous CO_2 injection at 13.0 ppg pressure gradient. Considerable narrow breakout shear failures and hydraulic fractures predicted during at ~7000-7400 ft MD. DoD defined as the "depth of damage".

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026

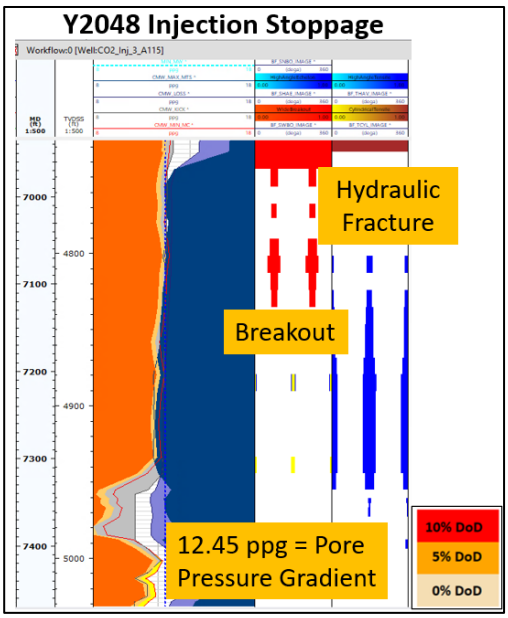


Fig 10: Injector I-3 in Y2048 during injection stoppage. Considerable wide breakouts and hydraulic fractures are predicted at ~6970 - 7120 ft MD and ~7070-7400 ft MD, respectively. DoD defined as the "depth of damage".

4. DISCUSSIONS

The results of the openhole stability analysis provide critical insights into the prediction of the mechanical behaviour of the openhole wellbore during CO_2 injection, particularly under operational conditions such as during continuous injection or injection stoppage. These findings highlight the dynamic nature of the stresses acting on the wellbore throughout the injection lifecycle and emphasise the importance of implementing suitable completions methods to ensure long-term storage and containment integrity.

Stress-induced failures and fracturing risks in openhole wellbore

The results showed that the wellbore is generally quite stable during the initial stage of Y2025 which includes post-drilling, when drilling mud is displaced with brine, and during the first year of continuous CO_2 injection. However, once CO_2 injection stops due to operational reasons, the predicted wellbore stability analysis showed considerable wide breakouts in certain sections. The 10.8 ppg pore pressure gradient input is used based on the predicted dynamic model which was the reservoir pressure in the first year of injection.

After five years of continuous CO_2 injection (Y2030), significant hydraulic fractures are predicted in the upper to middle section of the reservoir. This is mainly contributed by the significant reduction in fracture gradient, caused by the thermal cooling after several years of injection. During this injection stage, the temperature reduction is estimated to be around 78 °F. Therefore, the hydraulic fractures are predicted to occur due to the 12.3 ppg injection pressure gradient exceeding the reduced fracture gradient in the reservoir. However, upon injection stoppage, both minor breakout and hydraulic fractures were observed in the minimum horizontal stress and maximum horizontal stress directions, respectively.

In the final time step (Y2048), reduction of temperature due to the CO_2 injection is the highest at 139 °F. This dramatic change in temperature further reduces the stresses, resulting in further reduction of the fracture gradient. Hence, during the continuous injection, significant wellbore failures were predicted throughout the entire reservoir. Both narrow breakout shear failures and hydraulic fractures were predicted in the maximum horizontal stress direction, as the 13.0 ppg pore pressure gradient exceeds the fracture gradient. However, during injection stoppage, although hydraulic fractures were still predicted, wide breakouts were also observed in the upper section of the reservoir in the minimum horizontal stress direction. The openhole stability of the wellbore is deemed to be the worst during this period, likely due to the large reduction in stresses caused by the cooling effects within the reservoir.

In general, even during continuous injection scenarios, narrow breakout shear failures can occur in the direction of the maximum horizontal stress, suggesting that the increase in pore pressure and associated reduction in effective stress can create an environment prone to shear failure due to excessively high wellbore pressure prior to hydraulic fracture (Tan and Willoughby, 1993). However, hydraulic fracturing is mainly predicted as a result of the elevated pore pressure during CO_2 injection (Rutqvist and Tsang, 2002 and Mathias et al., 2009). This can be particularly concerning because hydraulic fractures, if uncontrolled, can extend beyond the target storage reservoir and potentially compromise the integrity of the overlying caprock (Goodarzi et al., 2012 and Vilarrasa et al., 2015). The initiation of hydraulic fractures near the wellbore is a recognised risk during CO_2 injection, especially in high-pressure scenarios where the injection pressure may exceed the fracture gradient of the formation which is reduced due to the thermal stress reduction in the reservoir (Vilarrasa et al., 2014). In the case of openhole completions, the risk is exacerbated due to the absence of mechanical barriers such as casing or liner.

Impact of Injection Stoppages on Wellbore Stability

A notable finding of the study is the increased risk of wellbore instability during injection stoppages. The modelling results revealed that during these operational stoppages, while hydraulic fractures are still present, the stress redistribution around the wellbore resulted in more severe wide breakouts compared to during continuous injection scenarios. The combined effects of pore pressure dissipation and cooling effects of CO_2 injection reduced the horizontal stresses, exacerbating the likelihood of

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026

shear failure in certain sections within the reservoir formation. The cyclic stress behaviour due to stress relaxation from injection stoppages pose risks to the structural integrity of the wellbore. Over time, the repeated cycles of injections and stoppages may weaken the formation, leading to progressive damage and potential collapse of the wellbore.

These results highlight the importance of considering operational contingencies such as injection stoppages in the design and management of CO_2 injection wells. In CCS projects, operational stoppages may occur due to various factors including maintenance, equipment failure, or regulatory requirements. Hence, it is important to adopt suitable well completion designs that can withstand the stress variations associated with these events (Bruno, 2001).

Mitigating wellbore instability with perforated liner completions

Given the potential risks of openhole completions during CO_2 injection, the project team decided to adopt perforated liner completions for the reservoir. This approach provides a more robust solution to mitigate wellbore failure, especially in the context of long-term CO_2 injection operations. The use of perforated liner completions offers several advantages over openhole completions which include:

- Enhanced mechanical support: Installation of a pipe reinforces the wellbore, reducing the risk of collapse and providing additional mechanical support to resist shear failures.
- Controlled flow paths: By implementing zonal isolation and perforating the pipe at selected sections, injection can be targeted at the most stable part of the reservoir, minimising potential leakage or unintended fluid migration.

Implication for CCS projects and future work

The findings of this study provided valuable input in the context of well completions design and operational risk management. While openhole completions may be appealing based on the benefits of injectivity and cost, the risk they pose especially during injection stoppages suggest that for this project, alternative completions methods should be considered in the CO_2 injection wells. This approach not only enhances wellbore stability but also improves the overall reliability of the CO_2 storage operation. Future research and field tests are required to further refine the use of coupled thermal dynamic-geomechanical models in predicting wellbore stability under a variety of operational conditions. Additionally, more work is required to optimise completions designs that can withstand the specific stress regimes during CO_2 storage operations. The integration of advanced monitoring technologies may also be critical in validating model predictions and to detect early signs of wellbore instability.

5. CONCLUSIONS

The results of the openhole stability analysis of the injector well underscore the critical risks associated with openhole completions during CO_2 injection. The effects of pore pressure change, along with thermal cooling can cause complex stress redistribution in both the far-field and near-wellbore.

By extracting the stress changes from the 3D coupled thermal dynamic-geomechanics model throughout the CO_2 injection period at the wellbore, analyses of the wellbore stability were conducted. The results, which predicted significant narrow breakouts and hydraulic fractures during continuous injection, and wide breakouts and hydraulic fractures during injection stoppages, highlighted the need for a more robust completion design. Therefore, a decision to adopt perforated liner completions was made, driven by the need to enhance wellbore stability and ensure the long-term success of the CO_2 injection operation. This approach will provide the necessary mechanical support to withstand the dynamic stress conditions that may occur throughout the injection lifecycle. At the same time, it also offers better control over the injection interval and minimising the risk of wellbore collapse.

Similar analysis is recommended for future CCS projects to ensure the completion method is aligned with the overarching goal of ensuring safe, long-term CO_2 storage in the reservoir, protecting both the environment and the integrity of the overall storage project.

6. REFERENCES

Bruno, M. S. (2001, September). Geomechanical analysis and decision analysis for mitigating compaction related casing damage. In SPE Annual Technical Conference and Exhibition (pp. SPE-71695). SPE.

Cappa, F., & Rutqvist, J. (2011). Modeling of coupled deformation and permeability evolution during fault reactivation induced by deep underground injection of CO2. International Journal of Greenhouse Gas Control, 5(2), 336-346.

Carey, J. W., Svec, R., Grigg, R., Lichtner, P. C., Zhang, J., & Crow, W. (2009). Wellbore integrity and CO2-brine flow along the casing-cement microannulus. Energy Procedia, 1(1), 3609-3615.

Fjaer, E., Holt, R. M., Horsrud, P., Raaen, A. M., and Risnes, R. (1992). Petroleum Related Rock Mechanics. 1st ed. Elsevier.

Global CCS Institute. (2022). The Global Status of CCS 2022. Global CCS Institute.

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026

International Energy Agency (IEA). (2023). World Energy Outlook 2023. IEA Publications.

Mathias, S. A., Hardisty, P. E., Trudell, M. R., & Zimmerman, R. W. (2009). Screening and selection of sites for CO2 sequestration based on pressure buildup. International Journal of Greenhouse gas control, 3(5), 577-585.

Metz, B., Davidson, O., De Coninck, H. C., Loos, M., & Meyer, L. (2005). IPCC Special Report On Carbon Dioxide Capture And Storage. Cambridge: Cambridge University Press.

Paterson, L., Lu, M., Connell, L. D., & Ennis-King, J. (2008, September). Numerical modeling of pressure and temperature profiles including phase transitions in carbon dioxide wells. In SPE Annual Technical Conference and Exhibition (pp. SPE-115946). SPE.

Pruess, K. (2008). On CO 2 fluid flow and heat transfer behavior in the subsurface, following leakage from a geologic storage reservoir. Environmental Geology, 54, 1677-1686.

Rutqvist, J., & Tsang, C. F. (2002). A study of caprock hydromechanical changes associated with CO 2-injection into a brine formation. Environmental Geology, 42, 296-305..

Rutqvist, J., Vasco, D. W., & Myer, L. (2010). Coupled reservoir-geomechanical analysis of CO2 injection and ground deformations at In Salah, Algeria. International Journal of Greenhouse Gas Control, 4(2), 225-230.

Tan, C. P., & Willoughby, D. R. (1993, October). Critical mud weight and risk contour plots for designing inclined wells. In SPE Annual Technical Conference and Exhibition (pp. SPE-26325). SPE.

Vilarrasa, V., Olivella, S., Carrera, J., & Rutqvist, J. (2014). Long term impacts of cold CO2 injection on the caprock integrity. International Journal of Greenhouse Gas Control, 24, 1-13.

Vilarrasa, V., Makhnenko, R. Y., & Laloui, L. (2015, October). Influence of poromechanical and thermal properties of the caprock on the safety of CO2 storage. In Second EAGE workshop on geomechanics and energy (pp. cp-466). European Association of Geoscientists & Engineers.

Downloaded from onepetro.org/armaigs/proceedings-pdf/IGS24/IGS24/ARMA-IGS-2024-0222/4428144/arma-igs-2024-0222.pdf by Universiti Teknologi Malaysia user on 31 October 2026